

2-Page Proposal for the NoSQL Project

1. Team members in alphabetical order of the last name:

- Bhondve Aakanksha Prashant
- Gadikar Akanksha Uday
- Mukherjee Prerana

2. Briefly and clearly describe the problem you want to work on.

- A simple application where we can explore artists, streams, music trends, tracks, and music rankings from different regions around the world

3. Which NoSQL database do you want to use?

- MongoDB

4. Which public dataset will you use? Provide the URL of the dataset. Does the dataset have at least 20 attributes and 5,000 samples?

- Dataset : Spotify Weekly Top 200 Songs Streaming Data
- URL: <https://www.kaggle.com/datasets/yelexa/spotify200>
- The dataset has 36 attributes and more than 5,000 samples.

5. Describe the N+1 non-trivial data analysis queries/tasks you plan to design and implement. Do you take Option 1 or 2 described in the NoSQL project description on Brightspace?

- Option 1 considers a dataset that includes:
 - Analyze the trends of the artists and tracks that were released in the particular year
 - Find the most popular track and the respective genre of the artist that was on Spotify Charts for the week
 - Analyze the previous week's rank of a track, compare with the current rank and show the corresponding artist name

- Analyze the top tracks, artists from different regions around the world
- Show the streams along with the track name in a particular week
- If possible, would try to implement prediction of next popular artist based on the overall streaming of the track/album/song for the same dataset

6. Will you implement a Web interface or GUI? Remember *that simply plotting graphs or tables, for example, in Jupyter Notebooks doesn't count*, as specified in the NoSQL project description. For more details, refer to the project description.

- Development of an application using GUI to implement queries and fetch required data

7. Briefly describe your milestones to finish the project this semester.

- Study the database, understand attributes, and perform queries.
- Finalize the design and integrate UI with database.